TEXAS INSTRUMENTS
CD54HC367, CD74HC367, CD54HCT367
CD74HCT367, CD54HC368, CD74HC368, CD74HCT368
SCHS181E – NOVEMBER 1997 – REVISED FEBRUARY 2022

CDx4HC367, CDx4HC368, CDx4HCT367, CD74HCT368 High-Speed CMOS Logic Hex Buffer/Line Driver, Three-State Non-Inverting and Inverting

1 Features

- Buffered inputs
- High current bus driver outputs
- Two independent three-state enable controls
- Typical propagation delay t_{PLH}, t_{PHL} = 8 ns at V_{CC} = 5 V, C_L =15 pF, T_A = 25°C
- Fanout (over temperature range)
  - Standard outputs: 10 LSTTL Loads
  - Bus driver outputs: 15 LSTTL Loads
- Wide operating temperature range: -55°C to 125°C
- Balanced propagation delay and transition times
- Significant power reduction compared to LSTTL Logic ICs
- HC Types
  - 2 V to 6 V operation
  - High noise immunity: N_{IL} = 30%, N_{IH} = 30% of V_{CC} at V_{CC} = 5 V
- HCT Types
  - 4.5 V to 5.5 V operation
  - Direct LSTTL input logic compatibility, V_{IL} = 0.8 V (Max), V_{IH} = 2 V (Min)
  - CMOS input compatibility, $I_I \leq 1$ µA at V_{OL}, V_{OH}

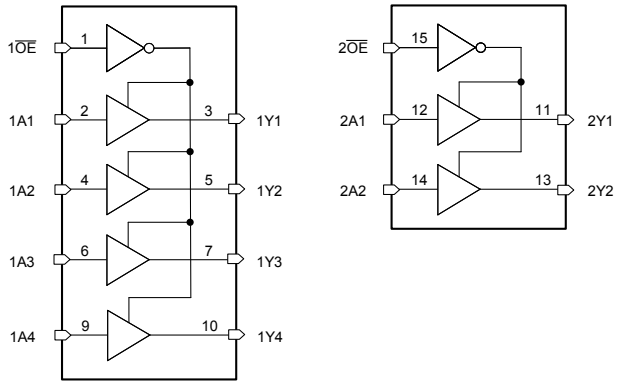
2 Description

The ’HC367, ’HCT367, ’HC368, and CD74HCT368 silicon gate CMOS three-state buffers are general purpose high-speed non-inverting and inverting buffers. The ’HC367 and ’HCT367 are non-inverting buffers, whereas the ’HC368 and CD74HCT368 are inverting buffers. They have high drive current outputs which enable high speed operation even when driving large bus capacitances. These circuits possess the low power dissipation of CMOS circuitry, yet have speeds comparable to low power Schottky TTL circuits. Both circuits are capable of driving up to 15 low power Schottky inputs.

Device Information

PART NUMBER	PACKAGE(1)	BODY SIZE (NOM)
CD74HC367M	SOIC (16)	9.90 mm × 3.90 mm
CD74HC368M	SOIC (16)	9.90 mm × 3.90 mm
CD74HCT367M	SOIC (16)	9.90 mm × 3.90 mm
CD74HCT368M	SOIC (16)	9.90 mm × 3.90 mm
CD74HC367E	PDIP (16)	19.31 mm × 6.35 mm
CD74HC368E	PDIP (16)	19.31 mm × 6.35 mm
CD74HCT367E	PDIP (16)	19.31 mm × 6.35 mm
CD74HCT368E	PDIP (16)	19.31 mm × 6.35 mm
CD54HC367F3A	CDIP (16)	24.38 mm × 6.92 mm
CD54HC368F3A	CDIP (16)	24.38 mm × 6.92 mm
CD54HCT367F3A	CDIP (16)	24.38 mm × 6.92 mm

(1) For all available packages, see the orderable addendum at the end of the data sheet.



Functional Block Diagram

An IMPORTANT NOTICE at the end of this data sheet addresses availability, warranty, changes, use in safety-critical applications, intellectual property matters and other important disclaimers. PRODUCTION DATA.

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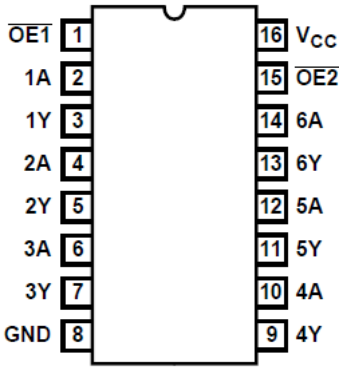
3 Revision History

NOTE: Page numbers for previous revisions may differ from page numbers in the current version.

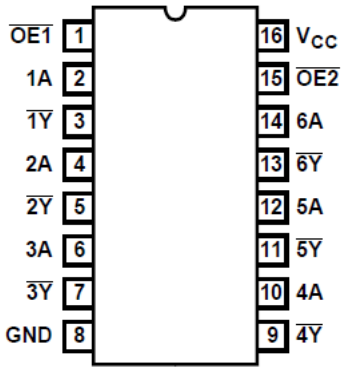
Changes from Revision D (October 2003) to Revision E (February 2022)	Page
• Updated the numbering, formatting, tables, figures, and cross-references throughout the document to reflect modern data sheet standards	1

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4 Pin Configuration and Functions



'HC367, 'HCT367
J, D, or N package
16-Pin CDIP, SOIC, PDIP
Top View



'HC368, CD74HCT368
J, D, or N package
16-Pin CDIP, SOIC, PDIP
Top View

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5 Specifications

5.1 Absolute Maximum Ratings

over operating free-air temperature range (unless otherwise noted)(1)

			MIN	MAX	UNIT
V_{CC}	Supply voltage range		-0.5	7	V
I_{IK}	Input clamp current	(V_I < -0.5 V or V_I > V_{CC} + 0.5 V)		±20	mA
I_{OK}	Output clamp current	(V_O < -0.5 V or V_O > V_{CC} + 0.5 V)		±20	mA
I_O	Continuous output current	(-0.5 V < V_O < V_{CC} + 0.5 V)		±35	mA
	Continuous current through V_{CC} or GND			±50	mA
T_J	Junction temperature			150	°C
T_{stg}	Storage temperature		-65	150	°C
	Lead Temperature (Soldering 10s)			300	°C

(1) Stresses beyond those listed under *absolute maximum ratings* may cause permanent damage to the device. These are stress ratings only and functional operation of the device at these or any other conditions beyond those indicated under *recommended operating conditions* is not implied. Exposure to absolute-maximum-rated conditions for extended periods may affect device reliability.

5.2 Recommended Operating Conditions

			MIN	MAX	UNIT
T_A	Temperature range		–55	125	℃
V_{CC}	Supply voltage range	HC Types	2	6	V
		HCT Types	4.5	5.5	V
V_I,V_O	Input or output voltage		0	V_{CC}	V
t_t	Input rise and fall time	2 V		1000	ns
		4.5 V		500	ns
		6 V		400	ns

5.3 Thermal Information

THERMAL METRIC		D (SOIC) 16 PINS	N (PDIP) 16 PINS	UNIT
$R_{\theta JA}$	Junction-to-ambient thermal resistance(1)	73	67	°C/W

(1) For more information about traditional and new thermal metrics, see the *Semiconductor and IC package thermal metrics* application report.

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5.4 Electrical Characteristics

PARAMETER		TEST CONDITIONS (2)	V_{CC} (V)	25℃			-40℃ to 85℃		-55℃ to 125℃		UNIT
				MIN	TYP	MAX	MIN	MAX	MIN	MAX	
HC TYPES											
V_{IH}	High level input voltage		2	1.5			1.5		1.5		V
			4.5	3.15			3.15		3.15		V
			6	4.2			4.2		4.2		V
V_{IL}	Low level input voltage		2			0.5		0.5		0.5	V
			4.5			1.35		1.35		1.35	V
			6			1.8		1.8		1.8	V
V_{OH}	High level output voltage	I_{OH} = – 20 μA	2	1.9			1.9		1.9		V
		I_{OH} = – 20 μA	4.5	4.4			4.4		4.4		V
		I_{OH} = – 20 μA	6	5.9			5.9		5.9		V
	High level output voltage	I_{OH} = – 6 mA	4.5	3.98			3.84		3.7		V
		I_{OH} = – 7.8 mA	6	5.48			5.34		5.2		V
V_{OL}	Low level output voltage	I_{OL} = 20 μA	2			0.1		0.1		0.1	V
		I_{OL} = 20 μA	4.5			0.1		0.1		0.1	V
		I_{OL} = 20 μA	6			0.1		0.1		0.1	V
	Low level output voltage	I_{OL} = 6 mA	4.5			0.26		0.33		0.4	V
		I_{OL} = 7.8 mA	6			0.26		0.33		0.4	V
I_I	Input leakage current		6			±0.1		±1		±1	μA
I_{CC}	Supply current	0	6			8		80		160	μA
I_{OZ}	Three-state leakage current	V_O = V_{CC} or GND	6			±0.5		±5.0		±10	μA
HCT TYPES											
V_{IH}	High level input voltage		4.5 to 5.5	2			2		2		V
V_{IL}	Low level input voltage		4.5 to 5.5			0.8		0.8		0.8	V
V_{OH}	High level output voltage	I_{OH} = – 20 μA	4.5	4.4			4.4		4.4		V
	High level output voltage	I_{OH} = – 4 mA	4.5	3.98			3.84		3.7		V
V_{OL}	Low level output voltage	I_{OL} = 20 μA	4.5			0.1		0.1		0.1	V
	Low level output voltage	I_{OL} = 4 mA	4.5			0.26		0.33		0.4	V
I_I	Input leakage current	V_I = V_{CC} to GND	5.5			±0.1		±1		±1	μA
I_{CC}	Supply current	V_I = V_{CC} to GND	5.5			8		80		160	μA
ΔI_{CC} (1)	Additional supply current per input pin	$\overline{OE1}$ input held at V_{CC} – 2.1	4.5 to 5.5		100	216		270		294	μA
		All other inputs held at V_{CC} – 2.1	4.5 to 5.5		100	198		247.5		269.5	
I_{OZ}	Three-state leakage current	V_O = V_{CC} or GND	5.5			±0.5		±5.0		±10	μA

(1) For dual-supply systems theoretical worst case (V_I = 2.4 V, V_{CC} = 5.5 V) specification is 1.8 mA.
(2) V_I = V_{IH} or V_{IL}, unless otherwise noted.

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5.5 Switching Characteristics

Input t_r, t_f = 6 ns. Unless otherwise specified, C_L = 50pF

PARAMETER		V_{CC} (V)	25℃		-40℃ to 85℃	-55℃ to 125℃	UNIT
			TYP	MAX	MAX	MAX	
HC TYPES							
t_{pd}	Data to outputs HC/HCT367	2		105	130	160	ns
		4.5		21	26	32	ns
		6	8(3)	18	24	27	ns
	Data to outputs HC/HCT368	2		105	130	160	ns
		4.5		21	26	32	ns
		6	9(3)	18	24	27	ns
	Output enable and disable to outputs	2		150	190	225	ns
		4.5		30	38	45	ns
		6	12(3)	26	33	38	ns
t_t	Output transition time	2		60	75	90	ns
		4.5		12	15	18	ns
		6		10	13	15	ns
C_I	Input capacitance			10	10	10	pF
C_O	Three-state output capacitance			20	20	20	pF
C_{PD}	Power dissipation capacitance(1) (2)	5	40				pF
HCT TYPES							
t_{pd}	Data to outputs HC/HCT367	4.5	9(3)	25	31	38	ns
	Data to outputs HC/HCT368	4.5	11(3)	30	38	45	ns
	Output enable and disable to outputs	4.5	14(3)	35	44	53	ns
t_t	Output transition time	4.5		12	15	18	ns
C_{IN}	Input capacitance			10	10	10	pF
C_O	Three-state capacitance			20	20	20	pF
C_{PD}	Power dissipation capacitance(1) (2)	5	42				pF

(1) C_{PD} is used to determine the dynamic power consumption, per buffer.
(2) $P_D = V_{CC}{}^2 f_i (C_{PD} + C_L)$ where f_i = input frequency, C_L = output load capacitance, V_{CC} = supply voltage.
(3) C_L = 15 pF and V_{CC} = 5 V.

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6 Parameter Measurement Information

t_{pd} is the maximum between t_{PLH} and t_{PHL}

t_t is the maximum between t_{TLH} and t_{THL}

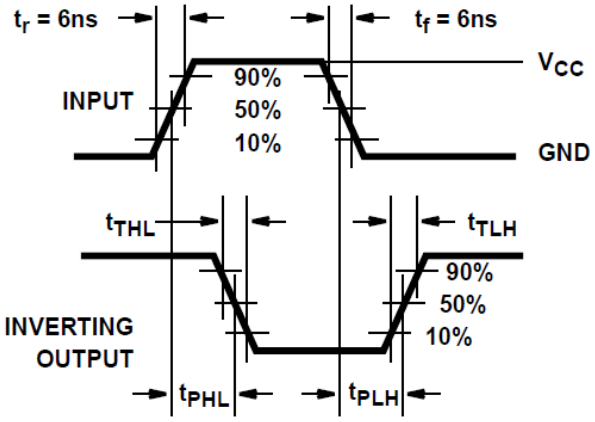


Figure 6-1. HC Transition Times and Propagation Delay Times, Combination Logic

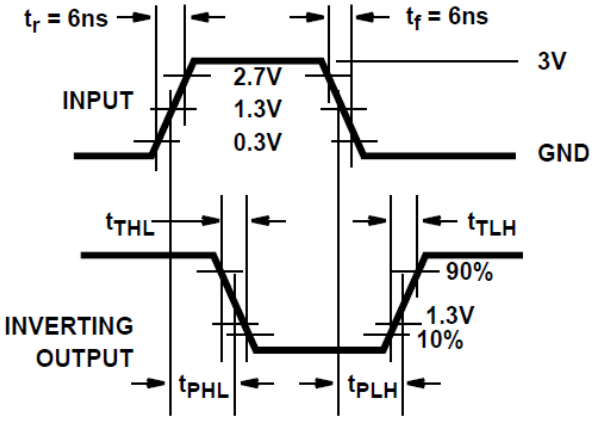


Figure 6-2. HCT Transition Times and Propagation Delay Times, Combination Logic

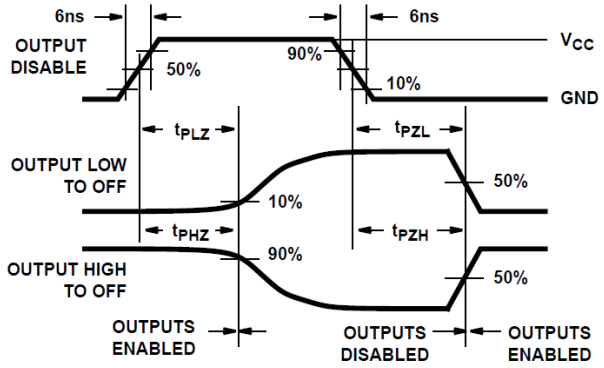


Figure 6-3. HC Three-State Propagation Delay Waveform

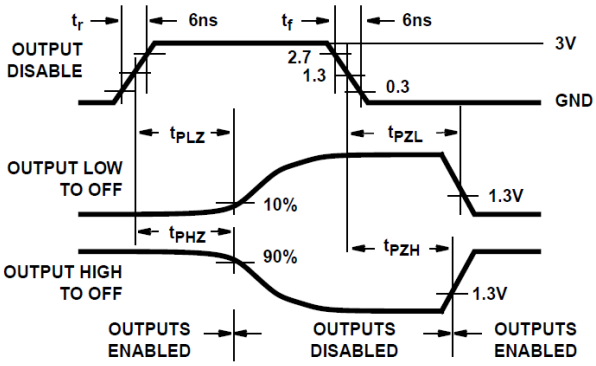
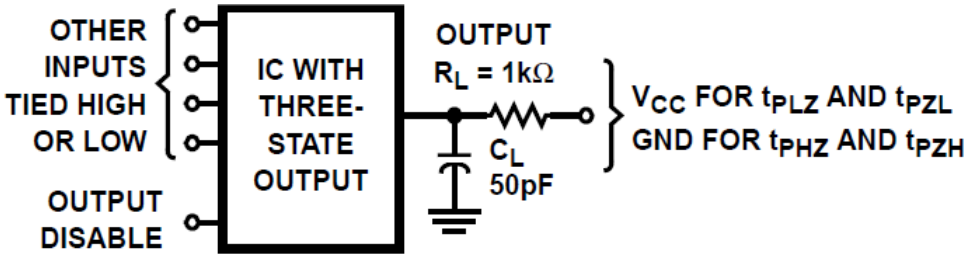


Figure 6-4. HCT Three-State Propagation Delay Waveform



Note

Open drain waveforms t_{PLZ} and t_{PZL} are the same as those for three-state shown on the left. The test circuit is Output R_L = 1 kΩ to V_{CC}, C_L = 50 pF.

Figure 6-5. HC and HCT Three-State Propagation Delay Test Circuit

7 Detailed Description

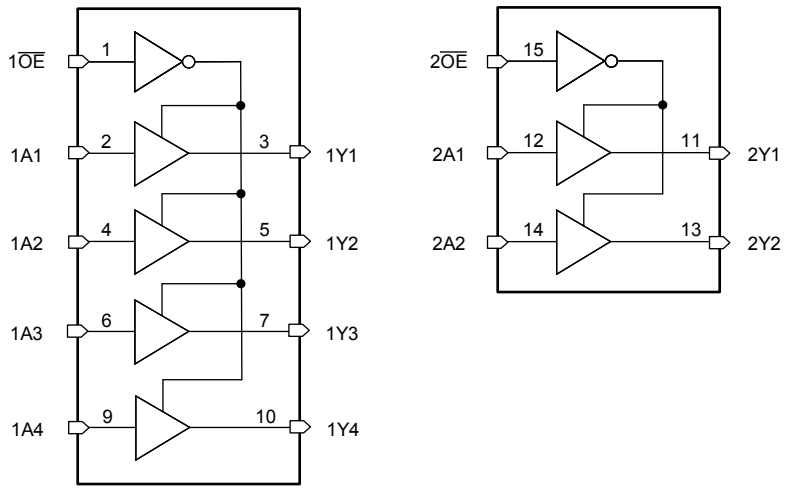
7.1 Overview

The ’HC367, ’HCT367, ’HC368, and CD74HCT368 silicon gate CMOS three-state buffers are general purpose high-speed non-inverting and inverting buffers. They have high drive current outputs which enable high speed operation even when driving large bus capacitances. These circuits possess the low power dissipation of CMOS circuitry, yet have speeds comparable to low power Schottky TTL circuits. Both circuits are capable of driving up to 15 low power Schottky inputs.

The ’HC367 and ’HCT367 are non-inverting buffers, whereas the ’HC368 and CD74HCT368 are inverting buffers. These devices have two output enables, one enable ($\overline{OE1}$) controls 4 gates and the other ($\overline{OE2}$) controls the remaining 2 gates.

The ’HCT367 and CD74HCT368 logic families are speed, function and pin compatible with the standard LS logic family.

7.2 Functional Block Diagram



7.3 Device Functional Modes

Table 7-1. Truth Table(1)

INPUTS		OUTPUTS (Y)	
$\overline{OE}$	A	HC/HCT367	HC/HCT368
L	L	L	H
L	H	H	L
H	X	(Z)	(Z)

(1) H = High Voltage Level, L = Low Voltage Level, X = Don’t Care, Z = High Impedance (OFF) State

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8 Power Supply Recommendations

The power supply can be any voltage between the minimum and maximum supply voltage rating located in the *Recommended Operating Conditions*. Each V_{CC} terminal should have a good bypass capacitor to prevent power disturbance. A 0.1-µF capacitor is recommended for this device. It is acceptable to parallel multiple bypass caps to reject different frequencies of noise. The 0.1-µF and 1-µF capacitors are commonly used in parallel. The bypass capacitor should be installed as close to the power terminal as possible for best results.

9 Layout

9.1 Layout Guidelines

When using multiple-input and multiple-channel logic devices inputs must not ever be left floating. In many cases, functions or parts of functions of digital logic devices are unused; for example, when only two inputs of a triple-input AND gate are used or only 3 of the 4 buffer gates are used. Such unused input pins must not be left unconnected because the undefined voltages at the outside connections result in undefined operational states. All unused inputs of digital logic devices must be connected to a logic high or logic low voltage, as defined by the input voltage specifications, to prevent them from floating. The logic level that must be applied to any particular unused input depends on the function of the device. Generally, the inputs are tied to GND or V_{CC}, whichever makes more sense for the logic function or is more convenient.

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10 Device and Documentation Support

TI offers an extensive line of development tools. Tools and software to evaluate the performance of the device, generate code, and develop solutions are listed below.

10.1 Receiving Notification of Documentation Updates

To receive notification of documentation updates, navigate to the device product folder on ti.com. Click on *Subscribe to updates* to register and receive a weekly digest of any product information that has changed. For change details, review the revision history included in any revised document.

10.2 Support Resources

TI E2E™ support forums are an engineer's go-to source for fast, verified answers and design help — straight from the experts. Search existing answers or ask your own question to get the quick design help you need.

Linked content is provided "AS IS" by the respective contributors. They do not constitute TI specifications and do not necessarily reflect TI's views; see TI's Terms of Use.

10.3 Trademarks

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10.4 Electrostatic Discharge Caution



This integrated circuit can be damaged by ESD. Texas Instruments recommends that all integrated circuits be handled with appropriate precautions. Failure to observe proper handling and installation procedures can cause damage.

ESD damage can range from subtle performance degradation to complete device failure. Precision integrated circuits may be more susceptible to damage because very small parametric changes could cause the device not to meet its published specifications.

10.5 Glossary

TI Glossary	This glossary lists and explains terms, acronyms, and definitions.

11 Mechanical, Packaging, and Orderable Information

The following pages include mechanical, packaging, and orderable information. This information is the most current data available for the designated devices. This data is subject to change without notice and revision of this document. For browser-based versions of this data sheet, refer to the left-hand navigation.

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PACKAGING INFORMATION

Orderable part number	Status (1)	Material type (2)	Package | Pins	Package qty | Carrier	RoHS (3)	Lead finish/ Ball material (4)	MSL rating/ Peak reflow (5)	Op temp (°C)	Part marking (6)
5962-9070601MEA	Active	Production	CDIP (J) | 16	25 | TUBE	No	SNPB	N/A for Pkg Type	-55 to 125	5962-9070601ME A CD54HCT367F3A
CD54HC367F3A	Active	Production	CDIP (J) | 16	25 | TUBE	No	SNPB	N/A for Pkg Type	-55 to 125	8500201EA CD54HC367F3A
CD54HC367F3A.A	Active	Production	CDIP (J) | 16	25 | TUBE	No	SNPB	N/A for Pkg Type	-55 to 125	8500201EA CD54HC367F3A
CD54HC368F3A	Active	Production	CDIP (J) | 16	25 | TUBE	No	SNPB	N/A for Pkg Type	-55 to 125	5962-8681201EA CD54HC368F3A
CD54HC368F3A.A	Active	Production	CDIP (J) | 16	25 | TUBE	No	SNPB	N/A for Pkg Type	-55 to 125	5962-8681201EA CD54HC368F3A
CD54HCT367F3A	Active	Production	CDIP (J) | 16	25 | TUBE	No	SNPB	N/A for Pkg Type	-55 to 125	5962-9070601ME A CD54HCT367F3A
CD54HCT367F3A.A	Active	Production	CDIP (J) | 16	25 | TUBE	No	SNPB	N/A for Pkg Type	-55 to 125	5962-9070601ME A CD54HCT367F3A
CD74HC367E	Active	Production	PDIP (N) | 16	25 | TUBE	Yes	NIPDAU	N/A for Pkg Type	-55 to 125	CD74HC367E
CD74HC367E.A	Active	Production	PDIP (N) | 16	25 | TUBE	Yes	NIPDAU	N/A for Pkg Type	-55 to 125	CD74HC367E
CD74HC367M	Obsolete	Production	SOIC (D) | 16	-	-	Call TI	Call TI	-55 to 125	HC367M
CD74HC367M96	Active	Production	SOIC (D) | 16	2500 | LARGE T&R	Yes	NIPDAU | SN	Level-1-260C-UNLIM	-55 to 125	HC367M
CD74HC367M96.A	Active	Production	SOIC (D) | 16	2500 | LARGE T&R	Yes	NIPDAU	Level-1-260C-UNLIM	-55 to 125	HC367M
CD74HC367M96G4	Active	Production	SOIC (D) | 16	2500 | LARGE T&R	Yes	NIPDAU	Level-1-260C-UNLIM	-55 to 125	HC367M
CD74HC367M96G4.A	Active	Production	SOIC (D) | 16	2500 | LARGE T&R	Yes	NIPDAU	Level-1-260C-UNLIM	-55 to 125	HC367M
CD74HC367MT	Obsolete	Production	SOIC (D) | 16	-	-	Call TI	Call TI	-55 to 125	HC367M
CD74HC368E	Last Time Buy	Production	PDIP (N) | 16	25 | TUBE	Yes	NIPDAU	N/A for Pkg Type	-55 to 125	CD74HC368E
CD74HC368E.A	Last Time Buy	Production	PDIP (N) | 16	25 | TUBE	Yes	NIPDAU	N/A for Pkg Type	-55 to 125	CD74HC368E
CD74HC368M	Active	Production	SOIC (D) | 16	40 | TUBE	Yes	NIPDAU	Level-1-260C-UNLIM	-55 to 125	HC368M
CD74HC368M.A	Active	Production	SOIC (D) | 16	40 | TUBE	Yes	NIPDAU	Level-1-260C-UNLIM	-55 to 125	HC368M
CD74HCT367E	Active	Production	PDIP (N) | 16	25 | TUBE	Yes	NIPDAU	N/A for Pkg Type	-55 to 125	CD74HCT367E

Orderable part number	Status (1)	Material type (2)	Package | Pins	Package qty | Carrier	RoHS (3)	Lead finish/ Ball material (4)	MSL rating/ Peak reflow (5)	Op temp (°C)	Part marking (6)
CD74HCT367E.A	Active	Production	PDIP (N) | 16	25 | TUBE	Yes	NIPDAU	N/A for Pkg Type	-55 to 125	CD74HCT367E
CD74HCT367M	Obsolete	Production	SOIC (D) | 16	-	-	Call TI	Call TI	-55 to 125	HCT367M
CD74HCT367M96	Active	Production	SOIC (D) | 16	2500 | LARGE T&R	Yes	NIPDAU | SN	Level-1-260C-UNLIM	-55 to 125	HCT367M
CD74HCT367M96.A	Active	Production	SOIC (D) | 16	2500 | LARGE T&R	Yes	NIPDAU	Level-1-260C-UNLIM	-55 to 125	HCT367M
CD74HCT367MT	Obsolete	Production	SOIC (D) | 16	-	-	Call TI	Call TI	-55 to 125	HCT367M
CD74HCT368E	Active	Production	PDIP (N) | 16	25 | TUBE	Yes	NIPDAU	N/A for Pkg Type	-55 to 125	CD74HCT368E
CD74HCT368E.A	Active	Production	PDIP (N) | 16	25 | TUBE	Yes	NIPDAU	N/A for Pkg Type	-55 to 125	CD74HCT368E
CD74HCT368M	Obsolete	Production	SOIC (D) | 16	-	-	Call TI	Call TI	-55 to 125	HCT368M
CD74HCT368M96	Active	Production	SOIC (D) | 16	2500 | LARGE T&R	Yes	NIPDAU	Level-1-260C-UNLIM	-55 to 125	HCT368M
CD74HCT368M96.A	Active	Production	SOIC (D) | 16	2500 | LARGE T&R	Yes	NIPDAU	Level-1-260C-UNLIM	-55 to 125	HCT368M
CD74HCT368MT	Obsolete	Production	SOIC (D) | 16	-	-	Call TI	Call TI	-55 to 125	HCT368M

(1) **Status:** For more details on status, see our product life cycle.

(2) **Material type:** When designated, preproduction parts are prototypes/experimental devices, and are not yet approved or released for full production. Testing and final process, including without limitation quality assurance, reliability performance testing, and/or process qualification, may not yet be complete, and this item is subject to further changes or possible discontinuation. If available for ordering, purchases will be subject to an additional waiver at checkout, and are intended for early internal evaluation purposes only. These items are sold without warranties of any kind.

(3) **RoHS values:** Yes, No, RoHS Exempt. See the TI RoHS Statement for additional information and value definition.

(4) **Lead finish/Ball material:** Parts may have multiple material finish options. Finish options are separated by a vertical ruled line. Lead finish/Ball material values may wrap to two lines if the finish value exceeds the maximum column width.

(5) **MSL rating/Peak reflow:** The moisture sensitivity level ratings and peak solder (reflow) temperatures. In the event that a part has multiple moisture sensitivity ratings, only the lowest level per JEDEC standards is shown. Refer to the shipping label for the actual reflow temperature that will be used to mount the part to the printed circuit board.

(6) **Part marking:** There may be an additional marking, which relates to the logo, the lot trace code information, or the environmental category of the part.

Multiple part markings will be inside parentheses. Only one part marking contained in parentheses and separated by a "~" will appear on a part. If a line is indented then it is a continuation of the previous line and the two combined represent the entire part marking for that device.

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and accurate information but may not have conducted destructive testing or chemical analysis on incoming materials and chemicals. TI and TI suppliers consider certain information to be proprietary, and thus CAS numbers and other limited information may not be available for release.

In no event shall TI's liability arising out of such information exceed the total purchase price of the TI part(s) at issue in this document sold by TI to Customer on an annual basis.

OTHER QUALIFIED VERSIONS OF CD54HC367, CD54HC368, CD54HCT367, CD74HC367, CD74HC368, CD74HCT367 :

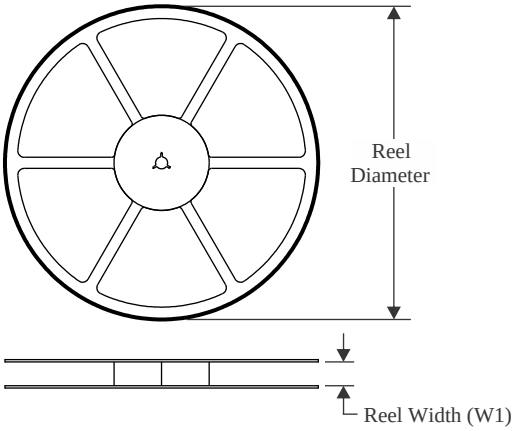
- Catalog : CD74HC367, CD74HC368, CD74HCT367
- Military : CD54HC367, CD54HC368, CD54HCT367

NOTE: Qualified Version Definitions:

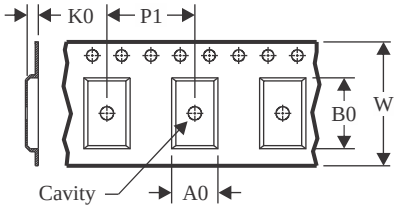
- Catalog - TI's standard catalog product
- Military - QML certified for Military and Defense Applications

TAPE AND REEL INFORMATION

REEL DIMENSIONS

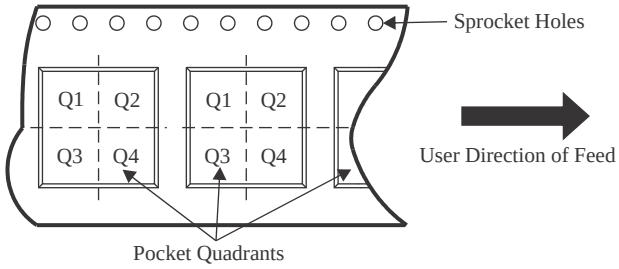


TAPE DIMENSIONS



A0	Dimension designed to accommodate the component width
B0	Dimension designed to accommodate the component length
K0	Dimension designed to accommodate the component thickness
W	Overall width of the carrier tape
P1	Pitch between successive cavity centers

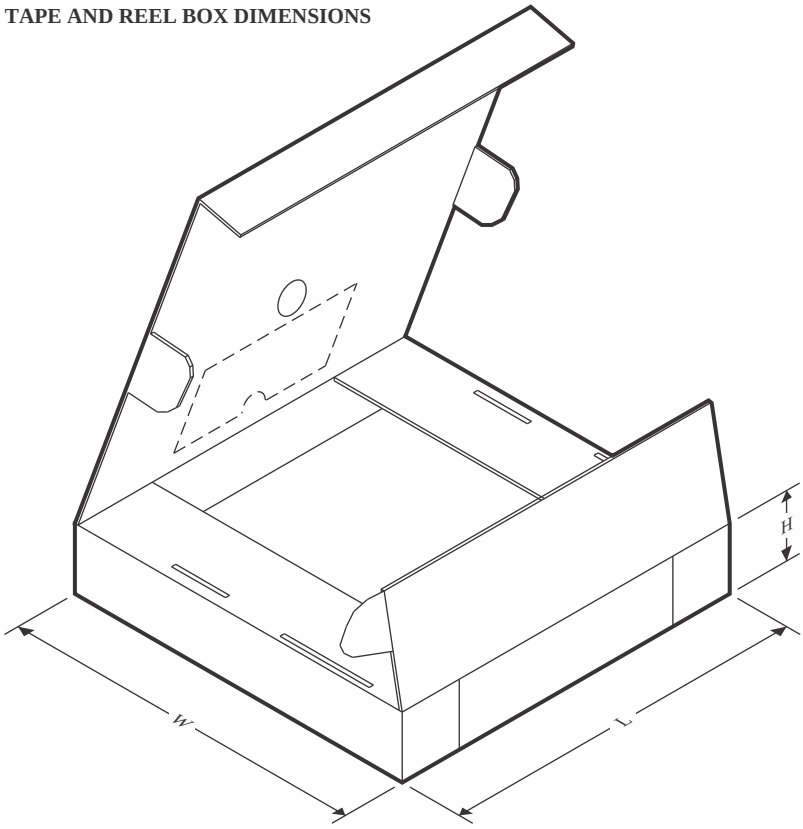
QUADRANT ASSIGNMENTS FOR PIN 1 ORIENTATION IN TAPE



*All dimensions are nominal

Device	Package Type	Package Drawing	Pins	SPQ	Reel Diameter (mm)	Reel Width W1 (mm)	A0 (mm)	B0 (mm)	K0 (mm)	P1 (mm)	W (mm)	Pin1 Quadrant
CD74HC367M96	SOIC	D	16	2500	330.0	16.4	6.5	10.3	2.1	8.0	16.0	Q1
CD74HC367M96G4	SOIC	D	16	2500	330.0	16.4	6.5	10.3	2.1	8.0	16.0	Q1
CD74HCT367M96	SOIC	D	16	2500	330.0	16.4	6.5	10.3	2.1	8.0	16.0	Q1
CD74HCT368M96	SOIC	D	16	2500	330.0	16.4	6.5	10.3	2.1	8.0	16.0	Q1

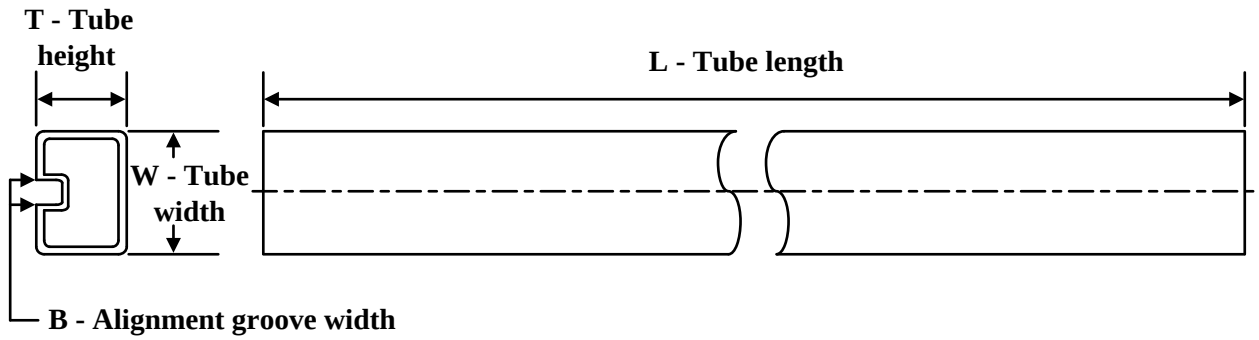
TAPE AND REEL BOX DIMENSIONS



*All dimensions are nominal

Device	Package Type	Package Drawing	Pins	SPQ	Length (mm)	Width (mm)	Height (mm)
CD74HC367M96	SOIC	D	16	2500	353.0	353.0	32.0
CD74HC367M96G4	SOIC	D	16	2500	353.0	353.0	32.0
CD74HCT367M96	SOIC	D	16	2500	353.0	353.0	32.0
CD74HCT368M96	SOIC	D	16	2500	353.0	353.0	32.0

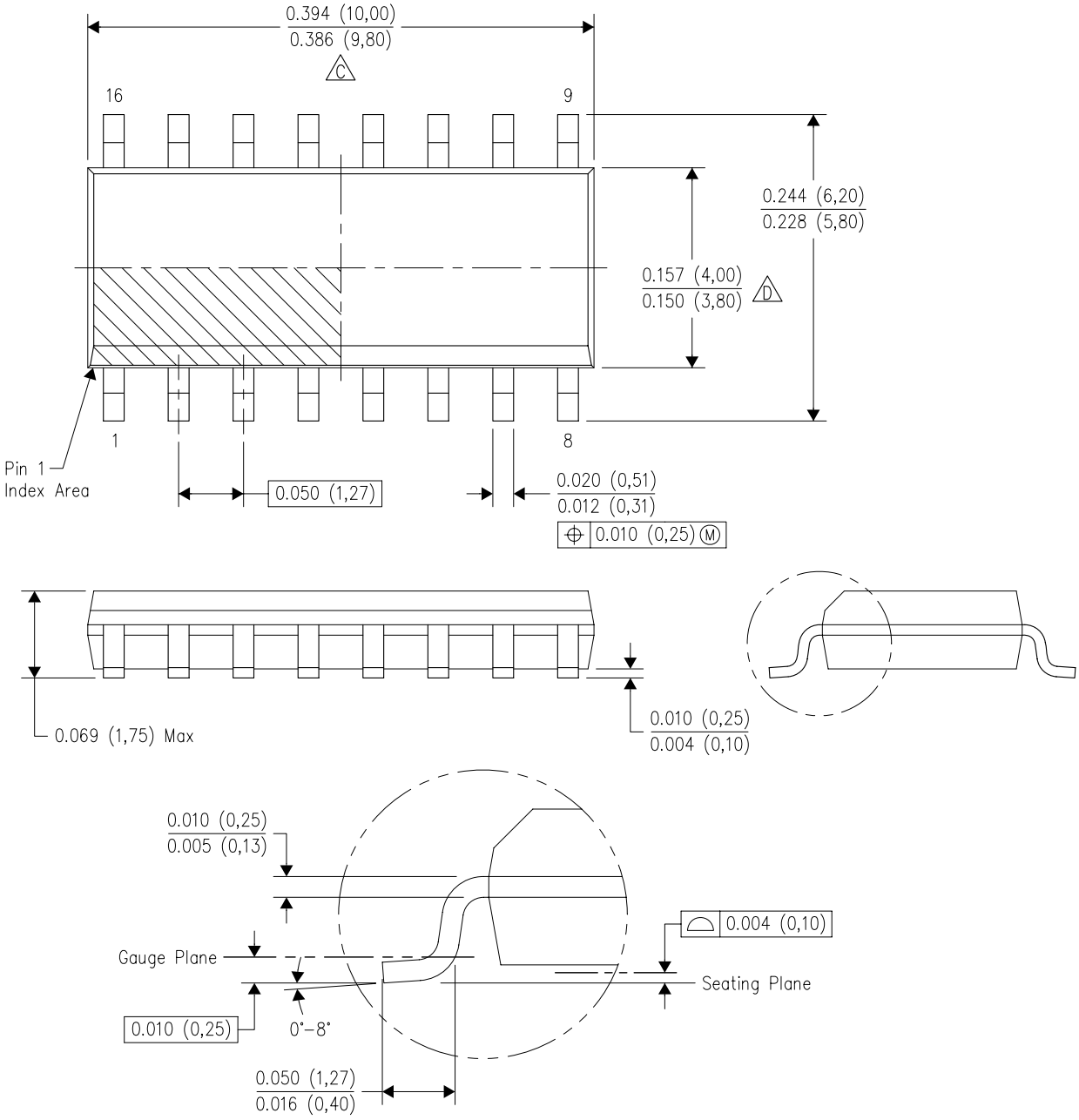
TUBE



*All dimensions are nominal

Device	Package Name	Package Type	Pins	SPQ	L (mm)	W (mm)	T (µm)	B (mm)
CD74HC367E	N	PDIP	16	25	506	13.97	11230	4.32
CD74HC367E	N	PDIP	16	25	506	13.97	11230	4.32
CD74HC367E.A	N	PDIP	16	25	506	13.97	11230	4.32
CD74HC367E.A	N	PDIP	16	25	506	13.97	11230	4.32
CD74HC368E	N	PDIP	16	25	506	13.97	11230	4.32
CD74HC368E	N	PDIP	16	25	506	13.97	11230	4.32
CD74HC368E.A	N	PDIP	16	25	506	13.97	11230	4.32
CD74HC368E.A	N	PDIP	16	25	506	13.97	11230	4.32
CD74HC368M	D	SOIC	16	40	507	8	3940	4.32
CD74HC368M.A	D	SOIC	16	40	507	8	3940	4.32
CD74HCT367E	N	PDIP	16	25	506	13.97	11230	4.32
CD74HCT367E	N	PDIP	16	25	506	13.97	11230	4.32
CD74HCT367E.A	N	PDIP	16	25	506	13.97	11230	4.32
CD74HCT367E.A	N	PDIP	16	25	506	13.97	11230	4.32
CD74HCT368E	N	PDIP	16	25	506	13.97	11230	4.32
CD74HCT368E	N	PDIP	16	25	506	13.97	11230	4.32
CD74HCT368E.A	N	PDIP	16	25	506	13.97	11230	4.32
CD74HCT368E.A	N	PDIP	16	25	506	13.97	11230	4.32

D (R–PDSO–G16) PLASTIC SMALL OUTLINE

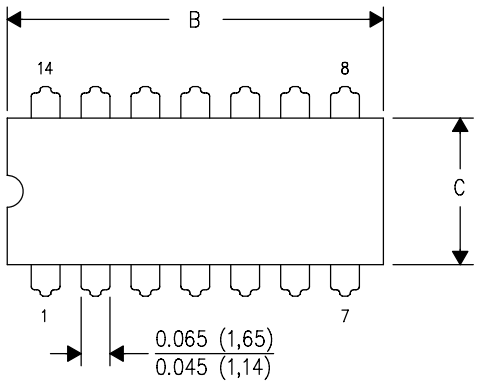


NOTES:
A. All linear dimensions are in inches (millimeters).
B. This drawing is subject to change without notice.
C. Body length does not include mold flash, protrusions, or gate burrs. Mold flash, protrusions, or gate burrs shall not exceed 0.006 (0,15) each side.
D. Body width does not include interlead flash. Interlead flash shall not exceed 0.017 (0,43) each side.
E. Reference JEDEC MS–012 variation AC.

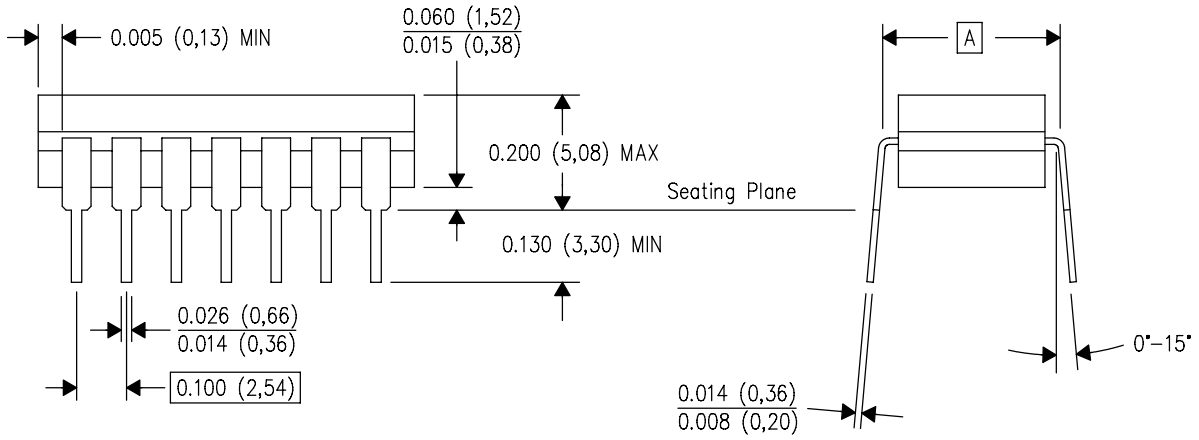
J (R–GDIP–T**)

14 LEADS SHOWN

CERAMIC DUAL IN–LINE PACKAGE



DIM \ PINS **	14	16	18	20
A	0.300 (7,62) BSC	0.300 (7,62) BSC	0.300 (7,62) BSC	0.300 (7,62) BSC
B MAX	0.785 (19,94)	.840 (21,34)	0.960 (24,38)	1.060 (26,92)
B MIN	—	—	—	—
C MAX	0.300 (7,62)	0.300 (7,62)	0.310 (7,87)	0.300 (7,62)
C MIN	0.245 (6,22)	0.245 (6,22)	0.220 (5,59)	0.245 (6,22)



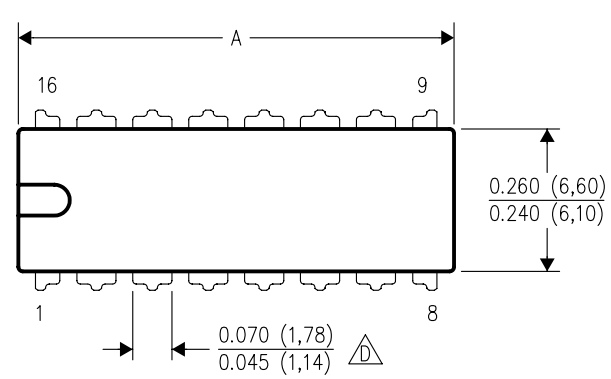
4040083/F 03/03

NOTES:
A. All linear dimensions are in inches (millimeters).
B. This drawing is subject to change without notice.
C. This package is hermetically sealed with a ceramic lid using glass frit.
D. Index point is provided on cap for terminal identification only on press ceramic glass frit seal only.
E. Falls within MIL STD 1835 GDIP1–T14, GDIP1–T16, GDIP1–T18 and GDIP1–T20.

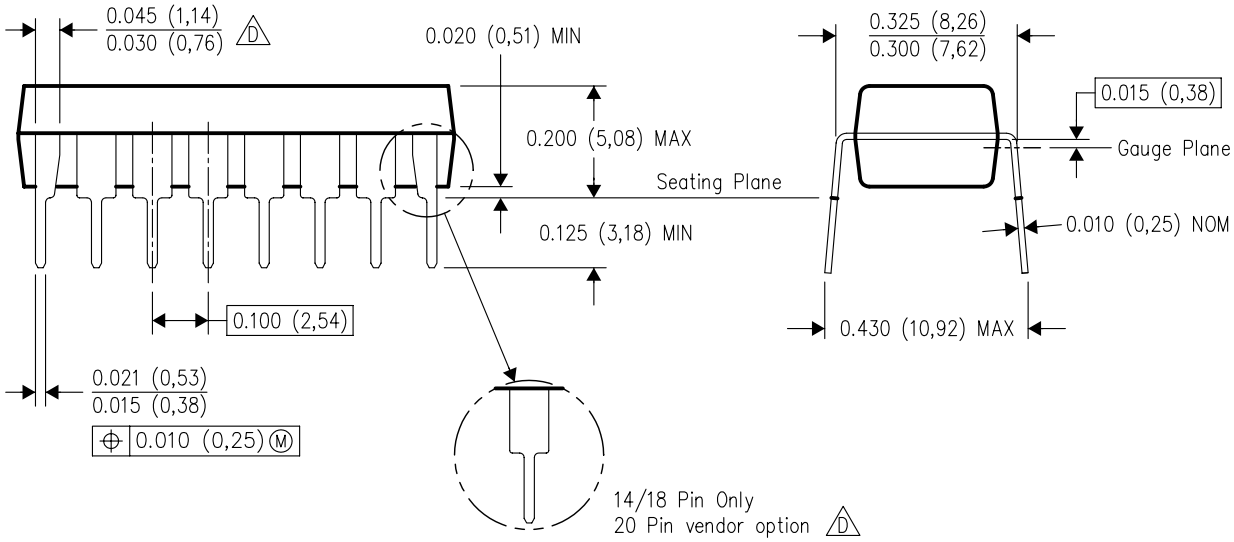
N (R–PDIP–T**)

PLASTIC DUAL–IN–LINE PACKAGE

16 PINS SHOWN



DIM \ PINS **	14	16	18	20
A MAX	0.775 (19,69)	0.775 (19,69)	0.920 (23,37)	1.060 (26,92)
A MIN	0.745 (18,92)	0.745 (18,92)	0.850 (21,59)	0.940 (23,88)
MS–001 VARIATION (C)	AA	BB	AC	AD



4040049/E 12/2002

NOTES:
A. All linear dimensions are in inches (millimeters).
B. This drawing is subject to change without notice.
C. Falls within JEDEC MS–001, except 18 and 20 pin minimum body length (Dim A).
D. The 20 pin end lead shoulder width is a vendor option, either half or full width.

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